MARWAN AL-MASRAT

Machine Learning Engineer

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Profile Summary

ML/AI Solutions Specialist with 1 year of experience in Deep Learning applications and advanced **NLP/LLM** solutions. Highly proficient in the technical orchestration of **MLOps** stages, with a proven track record of building deployable production systems, such as advanced **RAG** frameworks, and a background working within the **AWS** environment. Expert in utilizing **Python** and **PyTorch/TensorFlow** to strategically deliver direct and effective business value.

Areas of Expertise:

Data Analysis, Machine Learning, Deep Learning, LLM, Natural Language Processing (NLP), tokenization, MLOps, LLM, SQL, n8n workflow.

Skills

- **Technical Skills:**
  - **Tools & Languages**: Python, SQL, Git, Power BI, Linux, VScode, AWS Background
  - **Packages:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, Keras, PyTorch, NLTK, spaCy, Jupyter, Streamlit, PyTorch.

Development

- **Computer Vision Based Object Detection for Jordanian Football**
  - Developed and fine-tuned YOLOv5 models to detect players, ball, referees, and goalkeepers in Jordanian football videos.
  - Annotated ~300 frames with Data Augmentation to adapt to local video conditions.
  - **Tech Stack:** Python, YOLOv5, Roboflow | GitHub Repo link.

- **AI RAG Assistant Using LangChain.**
  - Engineered an AI-powered question-answering system that extracts precise answers from PDF documents using Retrieval-Augmented Generation. Integrated vector-based semantic search with Large Language Models to deliver real-time, context-aware responses.
  - **Tech Stack:** LangChain , RAG Architecture, ChromaDB, IBM Watsonx AI, Python

- **GenAI_using Reasoning &Prompt Engineer.**
  - Built a pipeline to analyze 9,400+ Yelp reviews, performing sentiment classification and extracting actionable business insights using a GenAI reasoning engine with prompt engineering.
  - Integrated OpenRouter API for multi-turn conversational analysis in Arabic and automated the workflow from raw feedback to structured recommendations.
  - **Tech Stack:** Python, NLP, Transformers, Generative AI, Prompt Engineering, API Integration
- Solar Power Generation Prediction System | Link: [https://solarpowerpredictor1.streamlit.app/]
  In this application, I deliberately used the Keras library to test it. This application measures solar energy production.

- AWS AI Practitioner Prep Questions | **Link**: [https://examapp-nerwggjtv7zc3n4tshywwd.streamlit.app/]
  In this project, I developed a practice exam application for the AWS certification to assess my own skills.

- **Spam Detection System (App)** | **Link:** [https://spampredict.streamlit.app]
  Developed and deployed a **Spam Detection** system using **NLP** techniques (**TF-IDF** and **Random Forest** within a Pipeline), with the frontend hosted on **Streamlit** .

- Smart News Analyzer
  Developed an application for text summarization and analysis using Transformer-based architectures and LSTM model.

Experience

Machine Learning & Deep Learning | Qafza Tech Dec 2024 – May 2025

- Analyzed mobile user behavior and built a Classification model to understand individual differences, utilizing Data Analysis and Feature Engineering to achieve excellent results.
- Developed a Regression model for Student Performance Factor analysis using Pipelines to streamline workflow and improve efficiency.

Data Scientist Intern | SHAI FOR AI Jul 2024 – Dec 2024

- Demonstrated strong foundational modeling by building **Classification** and **Regression** models (**Scikit-Learn, Random Forests**) for prediction challenges.
- Advanced performance via **Ensemble** and **Deep Learning** models (**XGBoost, TensorFlow**) on complex classification tasks.
- Conducted in-depth **Exploratory Data Analysis (EDA)** and statistical inference on large, complex datasets (e.g., Olympic history) to extract actionable insights.

Certifications

- The Complete SQL Bootcamp: Go from Zero to Hero from (Udemy).
- Project: Generative AI Applications with RAG and LangChain
- Fundamentals of AI Agents Using RAG and LangChain.
- AI Agents Using RAG and LangChain.
- Microsoft Power BI Desktop for Business Intelligence (Udemy).
- Introduction to Deep Learning & Neural Networks with Keras.
- Data Science Methodology from (Coursera).
- Python For Machine Learning & Data Science Masterclass from (Udemy).
- NLP - Natural Language Processing with Python (Udemy)
- How Transformer LLMs Work.
- Machine Learning in Production from Coursera.

Education

Bachelor's degree at Al-Hussein Bin Talal University **|** 2018–2021
Major: Physics **|** Grade: Very good

Achievements

- Founder and led a team (Sufra Alqalam) of 110 member and the Jordanian Ministry of Culture award a team with of 5,000 JOD in revenue for team.